

Total No. of Questions : 4]

SEAT No. :

PE-346

[Total No. of Pages : 2

[6580]-730

B.E. (Mechanical Honors) (Insem.)

**402034 MJ : MODELLING AND SIMULATION OF EHV
(2019 Pattern) (Semester - VII)**

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) Answer Q.1 or Q.2 and Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) Compare any two electric motor drives on basis of following [6]

- i) Efficiency
- ii) Speed
- iii) Size
- iv) Reliability
- v) Control stability
- vi) Performance

b) Explain Induction motor drives with its working principal, advantages, disadvantages and applications? [8]

OR

Q2) a) List and explain motor selection criteria for electric vehicles? [6]

b) What do you mean by EV configuration, explain with suitable block diagram? [8]

P.T.O.

Q3) a) Explain the constructional details of Ni-Cd battery with its advantages, disadvantages and applications? **[8]**

b) Explain different types of battery cooling with its significance? **[8]**

OR

Q4) a) Explain noise factors with its main sources and classification? **[8]**

b) Explain the Modal Analysis / Mode Shapes of in battery analysis with its experimental steps? **[8]**

